

- 2 A programmer uses a number of Abstract Data Types (ADT) in the programs that she is developing.

- (a) A stack using memory locations 400 to 409 is used in one program.

The diagram represents the current state of the stack.

A variable `TopOfStack` pointer indicates the last value added to the stack.

**Stack**

| Memory location | Value                         |
|-----------------|-------------------------------|
| 409             |                               |
| 408             |                               |
| 407             |                               |
| 406             |                               |
| 405             |                               |
| 404             |                               |
| 403             | 'P' ← <code>TopOfStack</code> |
| 402             | 'X'                           |
| 401             | 'D'                           |
| 400             | 'B'                           |

- (i) Complete the answer column in the following table:

|                                                                                                                          | Answer |
|--------------------------------------------------------------------------------------------------------------------------|--------|
| the memory location of the value that has been on the stack for the longest time                                         |        |
| the number of consecutive <b>push</b> operations that will result in the <code>TopOfStack</code> variable containing 409 |        |

[2]

